Question 1

1. HTTP two request methods
- GET
- POST

2. Client-Slide Web technologies
HTML
- Stands for Hypertext Markup Language
- To design the layout and structure of web pages

CSS
- Stands for Cascading Style Sheet
- To design the appearance and style of web pages.

JS
- Stands for JavaScript
- To design the frontend behavior and logic of web pages

PHP
- Stands for PHP Hypertext Preprocessor
- To design the backend behavior and logic of web pages

3. Web Client VS Web Server
Web Client
- Runs client-side scripts, javascript
- Requests and displays content from a web server
- Requests and renders web pages, resources and data from servers

Web Server
- runs server-side scripts, php
- stores, processes and delivers web pages to clients
- servers web pages, resources and data to clients

4. Server-side dynamic web pages
- Processing is performed at server-side
- Technology is PHP
- Server-side programming technology is more powerful

5. Client-side dynamic web pages
- Processing is performed at client-side
- Technology is JavaScript
- Client-side programming technology is less powerful

6. Input type #TODO

username<input type=”text”> // default(textbox) 单行input only

password<input type=”password”> // Password field

search<input type=”search”> // Search field

Slider<input type=”range”> // 滑动progress的

Spinner<input type=”number”> // 给user打number的

Radio button<input type=”radio”> // 单选button / Gender

Checkbox<input type=”checkbox”> // 当只有一个button，只能选一个，用在agree&not agree

Checkbox List<input type=”checkbox”> // 当有多个utton，能选完一或多

File picker<input type=”file”> // File uploads
根据题目给的时间放对应的下去
<input type=”???”>
datetime-local = Datetime picker
date = Date picker
time = Time picker
month = Month picker
week = Week picker

Q2

7. CSS - Inline and embedded

Inline	Embedded
CSS rules are added to a single Element via the style attribute	CSS rules are added to a single page via the <style> element
Affecting only this particular element	Affecting all element in this particular page
No reusability	Page-level reusability
Exp : <p style=”color:red”>Hello</p>	Exp : <style> p { color : red;} </style>

#HTML part#

8. HTML code - Form Input control

refers to Chapter 2C

```
<form>

<label>name</label><input type=”text” name=”name” id=”name>

<select>
        <option>abc</option>
</select>

<button type="submit">Submit</button>
<button type="reset">Reset</button>
<button type="button">Button</button>
```

</form>

HTML such as <h3>, <img>, <address> and <a>

9. HTML code - Table

Table - <table>
Table row - <tr>
Table header - <th>
Table data - <td>

#只能一行一行的放(row),不能一列一列将的(column)
#rowspan == 一行一行占位，往下的
#colspan == 一列一列占位，往旁边的

Case 1

Table 1 Participants Information

No.	Participant Name	Gender	Workshop Info	
			Date	Session
1	Ali bin Ahmad	Male	2025-11-25	Morning
2	Tracey Lim	Female	2025-11-26	Afternoon

```
<table border="1"> // 边框线
 <tr> // 第一行
   <th rowspan="2">No.</th><th rowspan="2">Participant
Name</th><th rowspan="2">Gender</th><th colspan="2">Workshop
Infor</th>
 </tr>
 <tr><th>Date</th><th>Session</th></tr> //第二行
 <tr> // 第四行, 上面的字已经撑大了, 下面的td跟着上面的, 不用set
colspan那些
  <td>1<td><td>Ali Bin
Ahmad<td><td>Male<td><td>2025-11-25<td><td>Morning<td>
 </tr>
 <tr> // 第五行
  <td>2<td><td>Trancy
Lim<td><td>Female</td><td>2025-11-26</td><td>Afternoon</td>
 </tr>
</table>
```

#自己要去多做练习

Q3

#JS part#

10. JS code - array and looping control

Exp :

```
fruits = [ ];
fruits[0] = ‘apple’;
fruits[1] = ‘banana’;

for(let i=0; i < fruits.length;i++){
    console.log(fruits[i]); }
```

11. JQuery selector categories

```
<img id=”img1” src=”#”>
<p class=”parag”>test</p>
<a href=”abc.com”>link</a>
```

#PHP part#

12. Different data type (three)

Scalar data types

- Scalar means data that contains a single value

Exp : Integer, float, string, boolean

Compound data types

- Compound means data that contains more than one value

Exp : Array, Object

Special data types

- Special means neither scalar nor compound but they have specific meaning

Exp : Resource, Null

13. String operator (read slides)

$s = ‘Hello World’;

- echo strlen(‘Hello World’); // 1
- echo $s[-1]; // ‘d’
- echo substr($s, 0, 5); // ‘Hello’ (start form index 0,read 5 char)
- echo substr($s, 0); // ‘Hello’

$s=’-XXX-XXX-XXX-’;

- echo substr_count($s, ‘-’); // 4
- echo substr_count($s, ‘X’); // 9

$s = ‘ABCDEF’;

- echo str_contains($s, ‘ ’) // true
- echo str_contains($s, ‘XX ’) // false
- echo str_starts_with($s, ‘AB ’) // true
- echo str_starts_with($s, ‘XX ’) // false
- echo strstr($s, ‘CD’) // ‘CDEF’

$s = AXBX

- echo str_replace(‘X’, ‘-’, $s, $n) // $n = A-B-

14. Associate array

Exp

```
$fruits = [
        ‘APR’ => ‘Apple’
        ‘BNN; => ‘Banana’
        ];
echo “ fruits[APR] and fruits[BNN] ”;
```

15. Superglobals

$_SERVER
- is a superglobal that stores information about the server and execution environment.
Exp : $method = $_SERVER['REQUEST_METHOD']

$_GET
- is a superglobal that stores inputs submitted via URL query string
Exp : $name = $_GET['name'] ?? ' ' ;

$_POST
- is a superglobal stores inputs submitted via POST request body
Exp : $name = $_POST['name] ?? ' ' ;

$_REQUEST
- is a superglobal that contains a combination of $_GET & $_POST
Exp : $name = $_REQUEST['name] ?? ' ' ;

Q4

#PHP Code + SQL statement#

16. PHP code to establish connection to MySQL database using PDO approach

Ans :

#database name = db

#username= root

#password = ‘ ’ (empty)

```
$_db = new PDO()’mysql:dbname=db’, ’root’, ’ ’, [
    PDO::ATTR_DEFAULT_FETCH_MODE => PDO::FETCH_OBJ,
] );
```

17. Statement to select record from DB (fetchAll , fetch, fetchColumn)

```
$stm = $_db -> query(‘SELECT * FROM fruit’);
$arr = $stm -> fetchAll(); // return all record
$arr = $stm -> fetch();     // return one record
$arr = $stm -> fetchColumn();  // return 1 column
```

18. DB query statement (SELECT AND INSERT)

(SELECT)

```
$stm = $_db ->prepare(‘SELECT * FROM fruit WHERE id=?’);
$stm->execute([$id]);
$fruit = $stm ->fetch();

# if要display result，create table的里面放这个来print
<table>
  <tr><th>Id</th><th>Fruit</th></tr>
  <tr>
     <?php foreach($fruit as $f) : ?>
        <td><?= $f[ ‘id’ ]?> </td>
        <td><?= $f[‘fruit’]?> </td>
     <?php end foreach; ?>
  </tr>
</table>
```

(INSERT)

```
$stm = $_db ->prepare(‘INSERT INTO fruit (name,price) VALUES
(?,?)’ );
```

#Theory part#

19. Advantages of using PDO (four)

- PDO uses object-oriented approach in coding
- PDO can work with different databases as long as the database-specific PDO drivers are available
- PDO providers a data-access abstraction layer
- Changing database only requires minimum rewrite to the database-specific SQL statements.

20. Authentication VS Authorization

Authentication	Authorization
Ensure if the user is really the one who he claims to be	Process of allowing or denying a user from accessing to protected pages
Know who is the user	Security access
Ask the user to provide a pair of email and password in login page	Only admin member can enter product maintenance page

Exp
// If login credentials are matched
$_SESSION['user_id'] = $id;
$_SESSION['user_role'] = $role;
	Exp
	$role = $_SESSION['user_role'] ?? null;
	if($role != 'Admin'){
	    header('Location:/login.php');
	    exit();
	}